Junior Makers · Class 12

Phone-Controlled RGB Mood Lamp

Change the colour of your lamp from a phone!

RGB colour mixing | Bluetooth HC-05 | MIT App Inventor



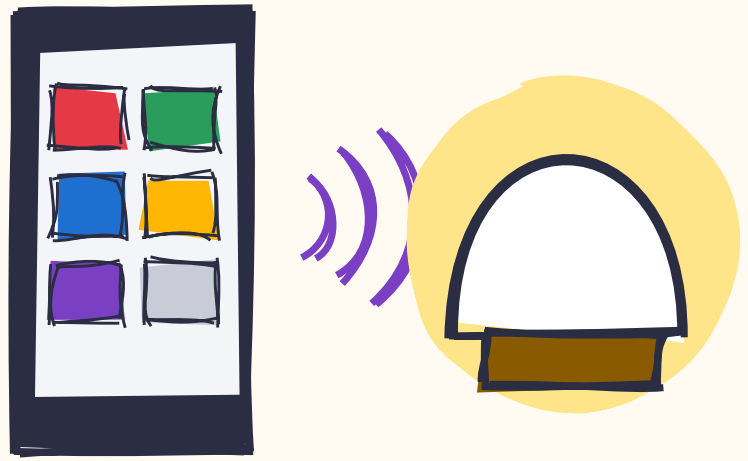
Did you know?
Your screen makes every colour you see from just three: red, green and blue.

Let's think...

How does a phone talk to a lamp?

No wires!

Smart bulbs change colour when you tap a phone. There is **no wire** between them. So how does the message get there?



Magic?

Through the air?

Radio waves!

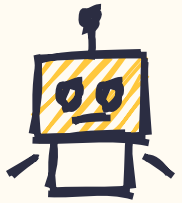
Today: we build a lamp that mixes any colour, and control it from an **Android phone** over Bluetooth.

Learning aims

Today I will learn to…

1. Mix colours with an **RGB LED** and PWM
2. Connect a **Bluetooth HC-05** module safely
3. Use an **App Inventor** app to send colour letters
4. Add a **goodnight tune** with the buzzer
5. Explain how devices send **small messages** wirelessly

Keywords
RGB · mix · Bluetooth · pair · message · app · command



Watch and think

Video time!



Colored Light Shadows: Additive Light Mixing

Homemade Science with Bruce Yeany · about 3 minutes · youtu.be/eKj1EwJ7THU

Watch for...

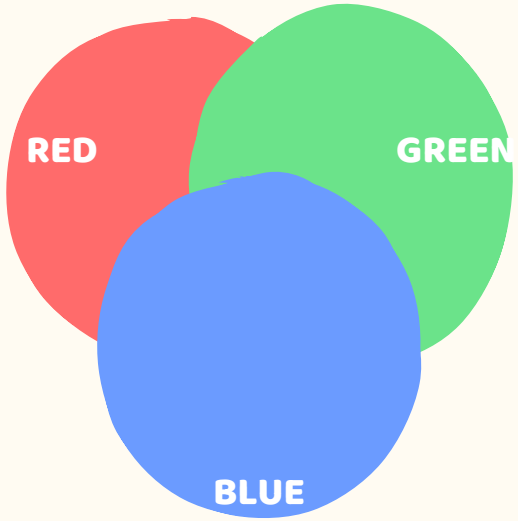
1. What colours of light does he start with?
2. What colour appears where **all three** lights overlap?
3. Why are the shadows different colours?

Scan to watch

After the video: Predict what colour red + green light makes. Test it on your RGB LED!

The science

Mixing LIGHT is not like mixing paint!



Light mixed	Colour you see
red + green	yellow
red + blue	magenta
green + blue	cyan
all three	**white**
none	off (black)

Paint vs light

Mix all paints and you get **muddy brown**. Mix all lights and you get **white**!

Look closely

Use a magnifying glass on a phone screen. You will see tiny red, green and blue dots!

Real world

Bluetooth is all around us



Smart bulbs
Change colour and brightness from an app



Earbuds
Music travels from phone to ears by Bluetooth



Game controllers
Send button presses to the console



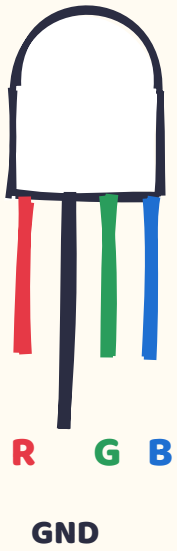
Smartwatches
Show phone messages on your wrist

Name a Bluetooth gadget in your home. What messages does it send?

Short range
Bluetooth works over about 10 metres, inside one room or house.

Meet the light

The RGB LED: 3 lights in 1



Inside the bulb

A **red**, a **green** and a **blue** LED share one bulb.
The **longest leg** is the common ground (GND).

Leg	Connect
R	pin 9 + 220 Ω
G	pin 10 + 220 Ω
B	pin 6 + 220 Ω
long leg	GND

Brightness 0 to 255

Each colour gets a number using **PWM**.
Orange = red 255, green 80, blue 0.

3 resistors

Each colour needs its OWN 220 Ω resistor, or one colour will hog the power.

Meet the radio

The HC-05 Bluetooth module



Pin	Connect
VCC	5V
GND	GND
TXD	pin 2 (Arduino listens)
RXD	pin 3 through a divider

3.3 V ears!

The HC-05 **talks** fine to the Arduino, but its **RXD** pin can be damaged by 5 V. Two resistors (**1k** and **2k**) make a **divider** that brings 5 V down to about 3.3 V.

Pairing code

When the phone asks: **1234** (sometimes 0000).

Android only

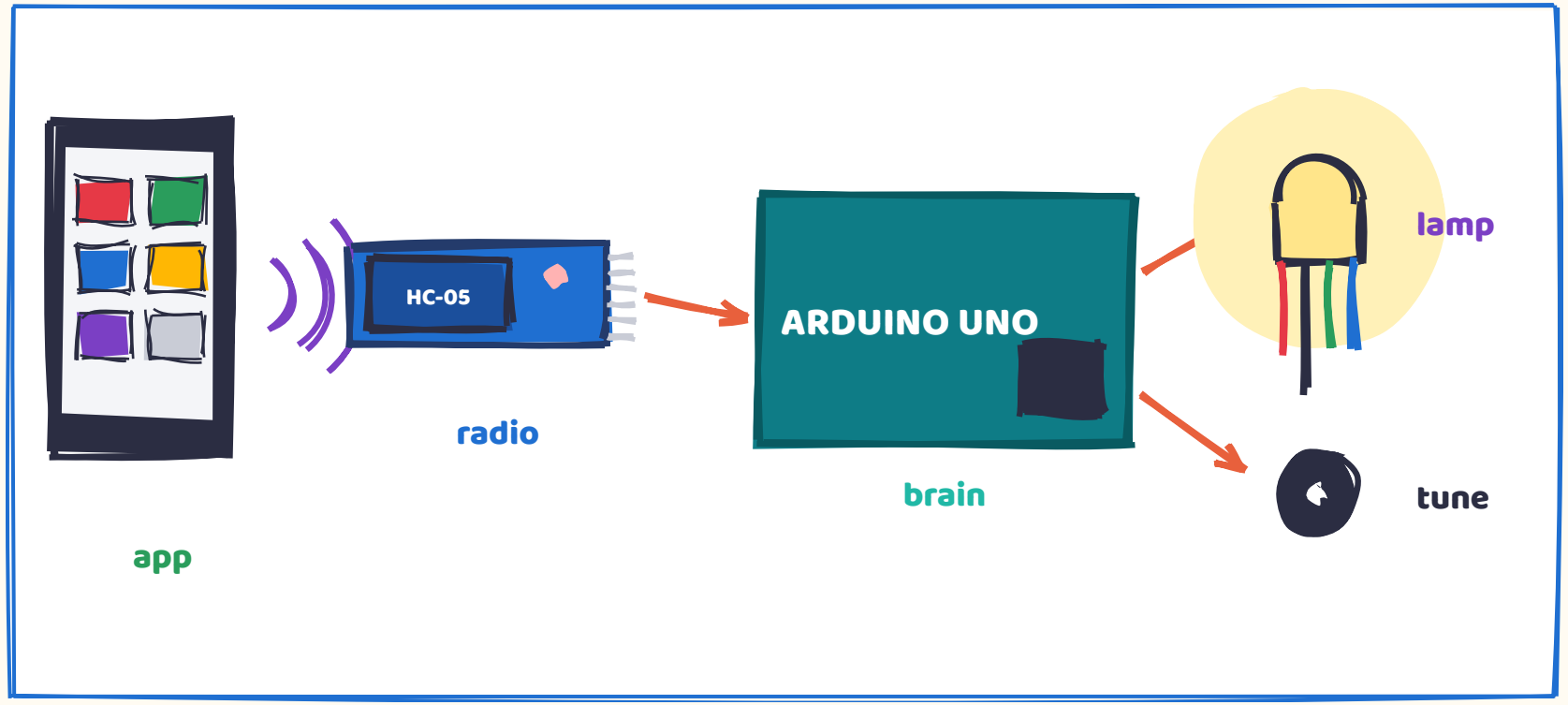
iPhones cannot talk to the HC-05. Use an Android phone or tablet.

LED blinks

Fast = waiting. Slow double blink = connected!

The plan

App, radio, brain, lamp



Two programs!

One runs on the **phone** (App Inventor).
One runs on the **Arduino** (ArduinoBlocks).
They share the letter code.

Team jobs

Builder · App designer · Tester. Swap next time!

Connect it

Pair the phone with the lamp

1. **Power** the Arduino. The HC-05 light blinks fast.
2. **Pair** in the phone's Bluetooth settings: choose HC-05, type 1234.
3. **Open** your app and press Connect. Choose HC-05 from the list.
4. **Check** the HC-05 light now blinks slowly: connected!
5. **Tap** a colour button. The lamp changes!

App can't find it?

On newer Android phones, allow **Nearby devices** (or Location) permission for the app in Settings.

Quick test

No app yet? Use the free Serial Bluetooth Terminal app and type R.

Part 2 · Make it

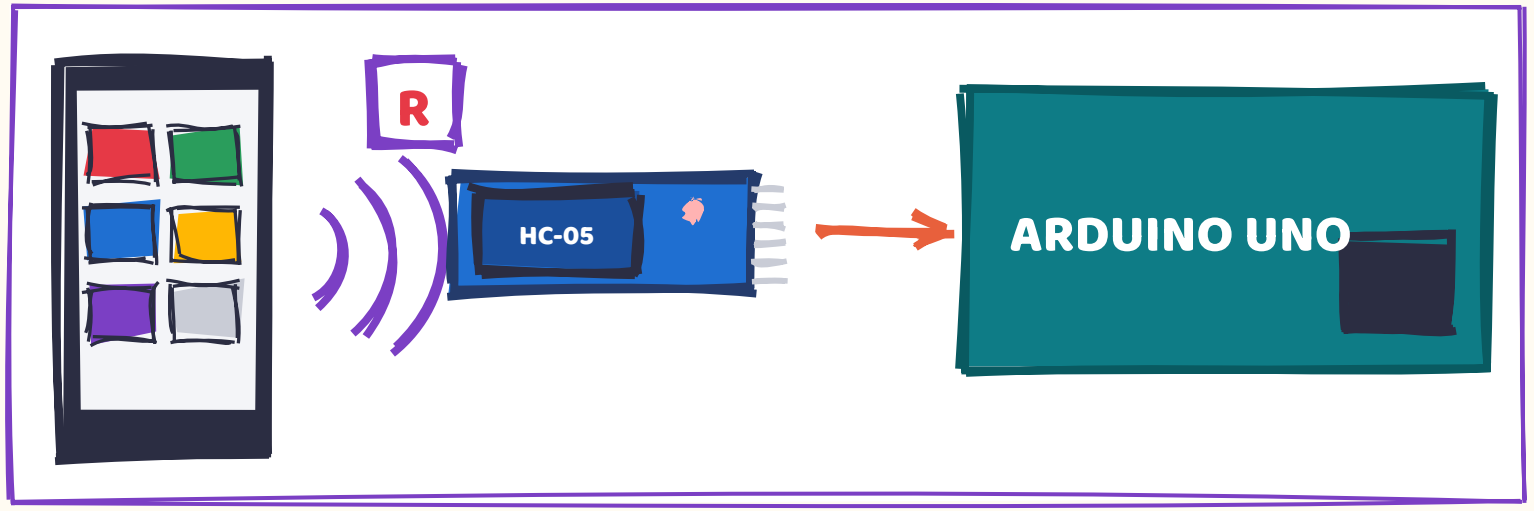
Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. The app sends ONE letter at a time
2. Be the lamp!
3. Our mood lamp kit (per group)
4. Wiring the mood lamp
5. The lamp in ArduinoBlocks
6. MIT App Inventor
7. Does every colour work?

Big idea

The app sends ONE letter at a time



Why letters?

A letter is a **tiny** message: quick to send and easy for the Arduino to check. Real smart bulbs also send short **commands**.

R	**G**	**B**	**Y**	**P**	**W**	**O**	**N**
red	green	blue	yellow	purple	white	off	goodnight

Same as...
Morse code in Little Explorers! A short code that means something bigger.

Class game

Be the lamp!

How to play

1. One student is the **phone**. Everyone else is a **lamp**.
2. Lamps hold up red, green and blue paper squares.
3. The phone calls a letter: **"Y!"**
4. Lamps hold up the right colours together (Y = red + green).
5. Wrong colour? Sit down!

Then discuss

What happens if the phone says a letter the lamp does not know, like **"Z"**?

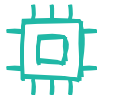


Big idea

Both sides must agree on the code BEFORE they talk. Engineers call this a protocol.

Build it

Our mood lamp kit (per group)



Arduino Uno
The brain



HC-05 module
Bluetooth radio



RGB LED
With 3 × 220 Ω



1k + 2k resistors
The divider



Buzzer
Goodnight tune



Android phone
With the app

Also: breadboard, jumper wires, USB cable, laptop, a paper cup or ping-pong ball as a lampshade.

Lampshade trick
Put a ping-pong ball over the LED. The colours mix much better!

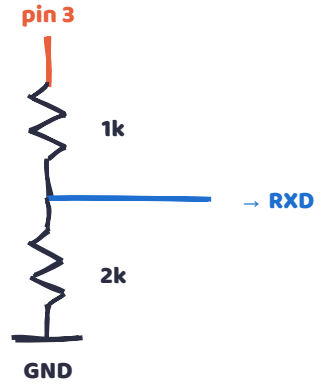
MAKE

Build it

Wiring the mood lamp

Part	Connect to
HC-05 VCC / GND	5V / GND
HC-05 TXD	pin 2
Arduino pin 3	1k → HC-05 RXD
HC-05 RXD	also 2k → GND
RGB red / green / blue	pins 9 / 10 / 6 (+ 220 Ω)
RGB long leg	GND
Buzzer +	pin 8

The divider



No 2k?

Use two 1k resistors in a line. 1k + 1k = 2k!

Why pins 2 and 3?

Pins 0 and 1 are used by the USB cable to upload code. Keep them free!

Code it

The lamp in ArduinoBlocks

```
when Arduino starts
    Bluetooth RX pin 2 · TX pin 3 · 9600 baud
forever
    if Bluetooth data available
        set msg to Bluetooth read
        if msg = R → RGB (255, 0, 0)
        if msg = G → RGB (0, 255, 0)
        if msg = B → RGB (0, 0, 255)
        if msg = Y → RGB (255, 180, 0)
        if msg = O → RGB (0, 0, 0)
        if msg = N → play goodnight tune, then RGB (0, 0, 0)
```

Read it like a story

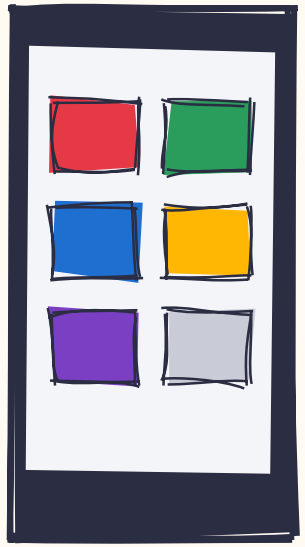
Wait for a letter.
Read it.
Check which letter it is.
Set the three colour numbers.



RGB (r, g, b)
means: analog write red pin 9, green pin 10, blue pin 6. Make a FUNCTION so you write it once!

The app

MIT App Inventor



Designer: what you see

A **ListPicker** button called Connect

One **Button** per colour

A hidden **BluetoothClient**

Ready-made app

The teacher shares a starter **.aia** file. Open it at **ai2.appinventor.mit.edu**, then test with the **MIT AI2 Companion** app.

```
when ListPicker.BeforePicking → Elements = BluetoothClient.AddressesAndNames
when ListPicker.AfterPicking → BluetoothClient.Connect (Selection)
when RedButton.Click → BluetoothClient.SendText "R"
```

Test like an engineer

Does every colour work?

Test	Expect	Result
Tap R	red	pass / fail
Tap G	green	pass / fail
Tap B	blue	pass / fail
Tap Y	yellow (red + green)	pass / fail
Tap N	tune, then off	pass / fail
Walk 10 m away and tap	?	pass / fail

Debug checklist

Wrong colours? Check which leg goes to which pin.
Nothing happens? Is the TXD wire on pin 2?
Upload fails? Unplug the HC-05 wires from pins 0 and 1.

Swap phones with another team. Can THEIR phone control YOUR lamp? Why could that be a problem?

Level up!

Mood lamp challenges

Choose your challenge

Your colour: add a new button and letter for your favourite colour (try pink: 255, 40, 120).

Bedtime mode: fade slowly from bright to off over 10 seconds.

Rainbow: a button that cycles through every colour.

Slider: use an App Inventor slider to send brightness numbers.

Think like an engineer

Anyone nearby with an app could connect to your lamp. How could you stop strangers controlling it?

Boss level

Change the HC-05 pairing code from 1234 to your own secret code.

Show what you know

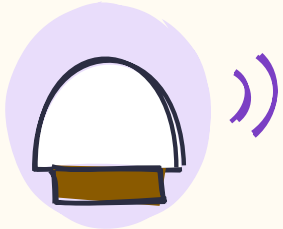
What did we learn?

- [x] Mixing red, green and blue light makes any colour
- [x] Each RGB colour gets a number from 0 to 255
- [x] Bluetooth sends messages by radio, no wires
- [x] Phone and lamp agree on a letter code (a protocol)
- [x] The HC-05 RXD pin needs a divider for 3.3 V

Exit ticket: which two colours of light make yellow?

At home

Find 3 things at home that use Bluetooth. Which one sends the most messages?



Next class: a Step Counter Fitness Tracker!